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Computer-readable recording medium, information processing method, and information processing apparatus

Abstract

A non-transitory computer-readable recording medium stores therein an information processing program that causes a computer to execute a process including, acquiring an image of an object storing items that is captured, extracting an object region of the object contained in the image by analyzing the image acquired, extracting one or more item regions contained in the extracted object region by analyzing the image acquired, estimating an item volume of the items stored in the object based on a position of an outline of the item region with respect to a position of the extracted object region, and specifying a number of items stored in the object based on the specified item volume.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S)

(1) This application is based upon and claims the benefit of priority of the prior Japanese Patent Application No. 2022-109953, filed on Jul. 7, 2022, the entire contents of which are incorporated herein by reference.

FIELD

(2) The embodiments discussed herein are related to a computer-readable recording medium, an information processing method, and an information processing apparatus.

BACKGROUND

(3) In a retail store, in order to eliminate long lines at the checkout, a system in which a customer personally register items by scanning the items and pays for the items is increasingly being introduced. Furthermore, in recent years, introduction of a system in which a customer scans items using an application that is installed in a terminal device that is lent in a retail store or a terminal device that the customer possesses has been started. In such a system in which a customer personally scans items, it is necessary to sense failing in scanning items in order to sense fraudulent behaviors, such as shoplifting, and errors.

(4) On the other hand, for example, a system that senses suspicious behaviors of customers and fraudulent behaviors, such as shoplifting, using a monitoring camera in a store is under development as a system that senses fraudulent behaviors of customers in a retail store. There is also a technique of sensing fraudulent behaviors and errors by specifying the number of items in a shopping basket from a video of a monitoring camera and comparing the specified number of items and items information on the items purchased by the customer with each other. Patent Literature 1: U.S. Pat. No. 8,448,858 Patent Literature 2: U.S. Patent Publication No. 2016/0189277 Patent Literature 3: Japanese Laid-open Patent Publication No. 2020-135422 Patent Literature 4: Japanese Laid-open Patent Publication No. 2013-041354

SUMMARY

(5) According to an aspect of an embodiment, a non-transitory computer-readable recording medium stores therein an information processing program that causes a computer to execute a process including, acquiring an image of an object storing items that is captured, extracting an object region of the object contained in the image by analyzing the image acquired, extracting one or more item regions contained in the extracted object region by analyzing the image acquired, estimating an item volume of the items stored in the object based on a position of an outline of the item region with respect to a position of the extracted object region, and specifying a number of items stored in the object based on the specified item volume.

(6) The object and advantages of the invention will be realized and attained by means of the elements and combinations particularly pointed out in the claims. It is to be understood that both

the foregoing general description and the following detailed description are exemplary and explanatory and are not restrictive of the invention, as claimed.

Description

BRIEF DESCRIPTION OF DRAWINGS

- (1) FIG. 1 is a diagram illustrating an example of purchasing items by self-scan;
- (2) FIG. 2 is a diagram illustrating an example of detecting items;
- (3) FIG. 3 is a diagram illustrating an example of a problem in detecting items; and
- (4) FIG. 4 is a diagram illustrating an example of a configuration of an information processing system according to Example 1;
- (5) FIG. 5 is a diagram illustrating an example of a configuration of an information processing device **10** according to Example 1;
- (6) FIG. 6 is a diagram illustrating an example of data that is stored in item information **32** according to Example 1;
- (7) FIG. 7 is a diagram illustrating an example of basket region rotational correction according to Example 1;
- (8) FIG. 8 is a diagram illustrating another example of the basket region rotational correction according to Example 1;
- (9) FIG. 9 is a diagram illustrating an example of estimating a volume of items in the basket according to Example 1;
- (10) FIG. 10 is a diagram illustrating another example of estimating a volume of items in the basket according to Example 1;
- (11) FIG. 11 is a diagram illustrating an example of a difference in how a basket and items are seen depending on the position of the basket according to Example 1;
- (12) FIG. 12 is a diagram illustrating an example of correcting a volume of items according to a position of a basket according to Example 1;
- (13) FIG. 13 is a flowchart illustrating an example of a flow of a number-of-items estimating process according to Example 1;
- (14) FIG. 14 is a flowchart illustrating another example of the flow of the number-of-items estimating process according to Example 1;
- (15) FIG. 15 is a flowchart illustrating still another example of the flow of the number-of-items estimating process according to Example 1;
- (16) FIG. 16 is a diagram illustrating an example of a hardware configuration of the information processing device **10**;
- (17) FIG. 17 is a diagram illustrating an example of a hardware configuration of an information processing terminal device **1000**; and
- (18) FIG. 18 is a diagram illustrating an example of a hardware configuration of a self-checkout terminal device **400**.

DESCRIPTION OF EMBODIMENTS

- (19) Preferred embodiments of the present invention will be explained with reference to accompanying drawings. Note that the examples do not limit the present embodiment and each example can be combined as appropriate within the scope without inconsistency.
- (20) First of all, using FIG. 1, a method enabling a customer to personally register items by scanning the items (sometimes “self-scan” below) and purchase the items will be described. FIG. 1 is a diagram illustrating an example of purchasing items by self-scan.
- (21) As illustrated in FIG. 1, first of all, a customer chooses a store that the customer is visiting via an application for registering items by scanning the items, which is an application displayed on a user terminal device **100**. The application is installed in the user terminal device **100** in advance.

The user terminal device **100** may be a mobile terminal device, such as a smartphone that the customer carries, or a dedicated terminal device that is lent to the customer in the store. In the case of the dedicated terminal device that is lent in the store, the customer need not choose a store and the store may be chosen in advance.

(22) The customer picks items to purchase and, for example, reads a barcode and a two-dimensional code of each item that is attached to the item or a store shelf, using the user terminal device **100** (sometimes referred to as “items-scan” below). Accordingly, the items to purchase are registered in the application.

(23) The customer then scans a payment code that is displayed on a display unit of a self-checkout terminal device **400**, or the like, using the user terminal device **100**. The customer then pays the amount that is displayed on a check-out screen on the self-checkout terminal device **400**, so that purchasing items completes.

(24) Purchasing items by self-scan has been described using FIG. 1. As for self-scan, however, for example, a customer can be spared from paying for items by putting the items in a basket without scanning the items and not making a payment with the self-checkout terminal device **400**.

Alternatively, a customer can be spared from paying for part of items to purchase in a way that the customer scans only part of the items and pays for only the scanned items at the self-checkout terminal device **400**. Particularly, fraudulent behaviors with a small number of items are easy for store staffs to find; however, for example, the case where the number of items is large and part of the items is not scanned is difficult for store staffs to find.

(25) There is thus a technique of specifying the number of items in a shopping basket from a video of a monitoring camera and comparing the specified number of items and item information on items that are purchased by a customer with each other, thereby sensing fraudulent behaviors and errors.

(26) FIG. 2 is a diagram illustrating an example of detecting items. As illustrated in FIG. 2, for example, it is possible to detect a shopping basket and items from a video of a monitoring camera, using an existing object detection algorithm. The existing object detection algorithm is, for example, an object detection algorithm using deep learning, such as Faster R-CNN (Regions with Convolutional Neural Network). Alternatively, the algorithm may be an object detection algorithm, such as YOLO (You Only Look Once) or SSD (Single Shot Multibox Detector) or DETR (DEtection TRansformer). Counting the number of the detected items in the shopping basket makes it possible to specify the number of items that are stored in the shopping basket.

(27) FIG. 2 exemplifies a video of a shopping basket that is shot from above, however, depending on the position of a monitoring camera or the shopping basket, a video may be shot from a side of the shopping basket or from diagonally above and thus there may be an item that is hidden behind the side of the shopping basket or another item and is not seen. In this case, an error may occur in specifying the number of items in a shopping basket from a video of the monitoring camera.

(28) FIG. 3 is a diagram illustrating an example of problems in detecting items. In the example on the left in FIG. 3, seven items are stored in a shopping basket but there are items that are hidden behind a side of the basket and, for example, the number of items may be specified as a number smaller than 7 according to a video that is shot from the side of the shopping basket. In the example on the right in FIG. 3, four items are stored in a shopping basket but there are items that are hidden behind other items and the number of items may be specified as a number smaller than 4. Thus, an object of the present embodiment is to estimate a volume of items in a basket in consideration of items that are hidden behind the side of the shopping basket or other items and specify the number of items in the shopping basket more accurately.

EXAMPLE 1

(29) First of all, an information processing system for carrying out the present embodiment will be described. FIG. 4 is a diagram illustrating an example of a configuration of an information processing system according to Example 1. As illustrated in FIG. 4, an information processing

system **1** is a system in which an information processing device **10** and user terminal devices **100-1** to **100-N** (collectively referred to as “a user terminal **100**” below and N is any integer) are connected via a network **50** such that mutual communications are enabled.

(30) The information processing device **10** is also connected to camera devices **200-1** and **200-2** (collectively referred to as “a camera device **200**” below) and a store staff terminal device **300** via the network **50** such that mutual communications are enabled. Furthermore, the information processing device **10** is also connected to self-checkout terminal devices **400-1** to **400-M** (collectively referred to as “a self-checkout terminal device **400**” below and M is an integer) via the network **50** such that mutual communications are enabled.

(31) For example, various communication networks, such as an intranet that is used in a retail store, can be employed as the network **50** regardless whether the networks are wired or wireless. The network **50** need not be a single network and, for example, may be configured in a way that an intranet and the Internet are configured via a network device, such as a gateway, and other devices (not illustrated in the drawing). Note that “in the retail store” is not limited to “in the building” and it covers “outside the building within the premises”.

(32) The information processing device **10** is set in, for example, a retail store and is an information processing device, such as a desktop personal computer (PC) or a laptop PC that is used by a store staff or a manager or a server computer.

(33) The information processing device **10**, for example, receives a plurality of captured images that are captured by the camera device **200** from the camera device **200**. Note that the captured images are a video that is shot by the camera device **200**, that is, a series of frames of a moving image.

(34) For example, using an existing object detection technique, the information processing device **10** detects a customer who stays in a store (sometimes simply referred to as “a person” below), a shopping basket (sometimes simply referred to as “a basket” below) that the person holds, and items from a captured image. As for detection of objects, such as items, from a captured image, for example, a given area on the captured image, such as a bounding box that is a rectangular region surrounding an object, such as an item, may be detected with respect to each object. For example, based on the basket and the items that are detected from the captured image, the information processing device **10** calculates the number of items that are stored in the basket.

(35) The information processing device **10** specifies the self-checkout terminal device **400** with which the person makes a payment, for example, based on position information on the person and position information on each of the self-checkout terminal devices **400-1** to **400-M**. Based on the difference between the number of items that are stored in the basket and the number of items to be purchased that are registered in the self-checkout terminal device **400**, the information processing device **10** determines whether there is an unregistered item that is not registered in the self-checkout terminal device **400** as an item to be purchased. When there is an unregistered item, the information processing device **10** notifies the store staff terminal device **300** of an alert.

(36) FIG. **4** illustrates the information processing device **10** as a single computer; however, the information processing device **10** may be a decentralized computing system consisting of a plurality of computers. Alternatively, the information processing device **10** may be a cloud computer device that is managed by a service provider that provides cloud computing services.

(37) The user terminal device **100** is, for example, an information processing terminal device for a customer to personally register an item to be purchased by scanning a barcode, or the like, of the item in order to purchase the item. The user terminal device **100** may be a mobile terminal device, such as a smartphone or a tablet PC that the customer owns, or may be a dedicated terminal device that is lent in the store. An application for registering items by scanning the items is installed previously in the user terminal device **100**.

(38) Using the user terminal device **100**, the customer, for example, registers items to be purchased in the user terminal device **100** by scanning a barcode of each item that is attached to the item or a

store shelf. When paying for the items, the customer, for example, causes the self-checkout terminal device **400** or a code reader (not illustrated in the drawing) that is set at the entry of a self-checkout area in which the self-checkout terminal device **400** is set to read a two-dimensional code that is displayed on the user terminal device **100**, or the like. Accordingly, item information on the items to be purchased is registered in the self-checkout terminal device **400** or the information processing device **10**.

(39) The camera device **200** is a monitoring camera that is set in a retail store or the premises of the retail store. The camera device **200** may include the camera device **200-1** for shooting a video of a shopping basket from above and the camera device **200-2** for shooting a video of the shopping basket from a side. Particularly when, a shopping cart on which shopping baskets can be placed on upper and lower two trays is used as illustrated in FIG. 4, an image of the shopping basket placed on the lower tray is shot by the camera device **200-2** that shoots a video from a side because it is difficult to shoot a video of the shopping basket on the lower tray with the camera device **200-1** that shoots a video from above. The camera device **200** may be set at the entry of the self-checkout area in which the self-checkout terminal device **400** is set or a place where shopping baskets are placed for each self-checkout terminal device **400**. A video that is shot by the camera device **200** is transmitted to the information processing device **10**.

(40) The store staff terminal device **300** may be a mobile terminal device, such as a smartphone or a tablet PC that a store staff of the retail store has, or may be an information processing device, such as a desktop PC or a laptop PC that is set in a given position in the store. When the information processing device **10** senses omission of scanning an item resulting from a fraudulent behavior or an error, the store staff terminal device **300** receives an alert from the information processing device **10**. Note that, for example, while there may be a plurality of the store staff terminal devices **300** for the respective store staffs in the store, the terminal device that is notified of the alert may be limited to a terminal that a shop staff who is in charge of security around the self-checkout area has.

(41) The self-checkout terminal device **400** is a self-checkout terminal device for a customer to personally make a payment to items. The self-checkout terminal device **400** receives information on items to be purchased that are registered via the user terminal device **100**. The self-checkout terminal device **400** receives a payment for the items to be purchased from the customer by cash, credit card, or cashless payment.

(42) Functional Configuration of Information Processing Device **10**

(43) A functional configuration of the information processing device **10** will be described next. FIG. 5 is a diagram illustrating an example of a configuration of the information processing device **10** according to Example 1. As illustrated in FIG. 5, the information processing device **10** includes a communication unit **20**, a storage unit **30**, and a controller **40**.

(44) The communication unit **20** is a processing unit that controls communications with other information processing devices, such as the user terminal device **100** and the camera device **200**. The communication unit **20** is, for example, a communication interface, such as a USB (Universal Serial Bus) interface or a network interface card.

(45) The storage unit **30** has a function of storing various types of data and a program that the controller **40** executes and the storage unit **30** is realized using, for example, a storage device, such as a memory or a hard disk. The storage unit **30** stores an image DB **31**, item information **32**, etc.

(46) The image DB **31** stores a plurality of captured images that are captured by the camera device **200** and that are a series of frames. The captured images that are stored in the image DB **31** are images that are captured by the camera device **200** and that are transmitted to the information processing device **10** and an identifier that uniquely identifies the camera device **200** that captures the image, the date at which the image is captured, etc., may be stored in the image DB **31**. The image DB **31** is capable of storing information on positions of objects, such as a person, a shopping basket and items, that are specified with respect to the captured image in the image.

(47) The item information **32** stores information on the items to be purchased that are registered by self-scan. FIG. **6** is a diagram illustrating an example of data that is stored in item information **32** according to Example 1. As illustrated in FIG. **6**, the item information **32** stores “item name” representing the name of an item that is registered by self-scan, “the number of items” representing the number of items per product, “price” representing the price of the item, etc., in association with one another. In addition to or instead of “item name”, “type of item” representing the type of the item that is registered by self-scan may be stored. An identifier of the captured image of the registered item can be stored in the item information **32** and can be linked with the captured image that is stored in the image DB **31**.

(48) The above-described information that is stored in the storage unit **30** is an example only and the storage unit **30** is capable storing various types of information in addition to the above-described information.

(49) The controller **40** is a processing unit that controls the entire information processing device **10** and is, for example, a processor. The controller **40** includes a detector **41**, a specifying unit **42**, a calculator **43**, and a notification unit **44**. Each of the processing units is an example of an electronic circuit that the processor includes or an example of a process that the processor executes.

(50) For example, by analyzing the image of the basket storing the items that is captured with the camera device **200**, the detector **41** extracts a basket region of the basket that is contained in the image. For example, by analyzing the image, the detector **41** extracts at least one item region that is contained in the extracted basket region. Note that extraction of the regions of the basket and item from the image may be performed using existing object detection, such as Faster R-CNN or YOLO.

(51) For example, when extracting the basket region, the detector **41** may execute rotational correction on the captured image of the basket to make a correction such that the sides of the basket are horizontal on the image. FIG. **7** is a diagram illustrating an example of basket region rotational correction according to Example 1. As illustrated in FIG. **7**, the detector **41** may extract the basket region after rotating the whole image by a given angle.

(52) FIG. **8** is a diagram illustrating another example of the basket region rotational correction according to Example 1. In the example according to FIG. **8**, the detector **41** rotates the image by a plurality of given angles, employs the smallest rectangular region that is detected, and extracts the employed rectangular region as the basket region.

(53) The specifying unit **42**, for example, specifies a position of an outline of an item region with respect to the basket region that is detected by the detector **41**. FIG. **9** is a diagram illustrating an example of estimating a volume of items in the basket according to Example 1. As illustrated in FIG. **9**, the item region that is extracted by the detector **41** may be a region of a part that is seen excluding the part hidden behind the basket.

(54) As illustrated on the left in FIG. **9**, the specifying unit **42** specifies a position of an outline of an item region **70** of an item with respect to a basket region **60** of the basket. This is, for example, as illustrated on the left in FIG. **9**, specifying a position of a top side of the outline of the item region **70** with a top side of the basket region **60** being set for 1 and a bottom side being set for 0. A no-item region **80** without an item is specified as the bottom side of the basket region **60**. The specifying unit **42**, for example, estimates an average of the top sides of the item region **70** and the no-item region **80** that is 0.2 in the example on the left in FIG. **9** as a volume of the item in the basket.

(55) As illustrated on the right in FIG. **9**, even when there are multiple items in the basket, as in the case illustrated on the left in FIG. **9**, the specifying unit **42** estimates an average of the top sides of each item region and a region without an item that is 0.8 in the example on the right in FIG. **9** as a volume of the items in the basket. Note that, as illustrated in FIG. **10**, also when the item regions **70** are extracted not by bounding boxes but by instance segmentation, the specifying unit **42** is capable of specifying a volume of the items in the basket based on positions of outlines of the item regions **70** with respect to the basket region **60**.

(56) FIG. 10 is a diagram illustrating another example of estimating a volume of items in the basket according to Example 1. For example, in the example according to FIG. 10, as illustrated at the center in FIG. 10, the specifying unit 42 masks and extracts the item regions 70 by instance segmentation. Even in this case, as illustrated on the right in FIG. 10, for example, an average of the top sides of the item regions 70 and the no-item region 80 can be estimated as a volume of the items.

(57) When estimating a volume of the items, the specifying unit 42 may correct and then estimate a volume of the items according to how the basket and the items are seen in the camera image. FIG. 11 is a diagram illustrating an example of a difference in how a basket and items are seen depending on a position of the basket according to Example 1. The three camera images on the left in FIG. 11 are images of the same basket placed on the lower tray of the shopping cart that are captured by the same camera device 200-2. As illustrated in FIG. 11, because the position of the shopping cart differs among these three camera images, how the basket and the items stored in the basket are seen differs. As illustrated on the right in FIG. 11, also when a basket region of the basket is extracted from each of the three camera images, how the basket and the items stored in the basket are seen differs. For this reason, a difference may occur also in the estimation of a volume of items described using FIGS. 9 and 10 depending on how the basket and the items stored in the basket are seen. For this reason, in the present embodiment, a volume of the items is corrected and then estimated according to how the basket and the items are seen in the camera images.

(58) FIG. 12 is a diagram illustrating an example of correcting a volume of items according to a position of a basket according to Example 1. The upper views in FIG. 12 present sets of a basket and items on which 0.65 and 0.5 that are amounts of the items are estimated, respectively, by the estimation of a volume of items described using FIGS. 9 and 10. The sets of a basket and items on which the amounts items of 0.65 and 0.5 are estimated are however the same set of a basket and items. The specifying unit 42 therefore corrects the volume of the items according to the position of the basket, for example, using Equation (1) below.

$$(59) \quad \alpha' = \frac{\alpha - k(x)}{1 - k(x)} \quad (1)$$

(60) In Equation (1), α denotes a volume of items before correction and α' denotes a volume of items after correction. Furthermore, x denotes a position of a basket that is, for example, a distance from a given reference to a center point of the basket in a camera image or a set of coordinates on the camera image. Furthermore, $k(x)$ denotes a correction value according to each position of the basket.

(61) Using Equation (1) corrects the position of the basket corresponding to the volume of items of 0, that is, the origin as the lower views in FIG. 12 present and accordingly the volume of the items is corrected. For this reason, also as for the values according to which the volumes of the items of 0.65 and 0.5 are estimated, corrections to volumes of the items of 0.3 and 0.3 are made, respectively. Note that, as described below, the volume of the items that is estimated by estimation of a volume of items like that illustrated using FIGS. 9 to 12 is then converted by the calculator 43 to the number of items.

(62) Back to description of FIG. 5, the specifying unit 42, for example, specifies a first self-checkout terminal device 400 that a first person with a basket uses from a second image of an area containing the self-checkout terminal device 400, which is captured with the camera device 200. The specifying unit 42, for example, receives item information on items to be purchased from the first self-checkout terminal device 400 and, based on the item information, specifies a third number of items to be purchased. The item information 32 that is received from the first self-checkout terminal device 400 is information that is generated in a way that the user terminal device 100 that is an information processing terminal device that the first person uses scans and then registers barcodes or two-dimensional codes that are assigned to the items.

- (63) Based on the number of items that is calculated based on the specified position of the outline of the item region like that illustrated using FIGS. 9 to 12 and the third number, the specifying unit 42 then determines whether there is an item that is not registered as an item to be purchased in the self-checkout terminal device 400.
- (64) The specifying unit 42 also specifies a personal item of the first person with the basket that are contained in the extracted basket region. To specify a personal item as described above, for example, a machine learning model that is generated using an existing machine learning technique by performing training such that a customer's own shopping bag or a bag is detected previously as a personal item from a captured image may be used. An object that is specified as a personal item is not detected as an item and is excluded from subjects on which an item region is extracted and a volume and the number of items are calculated.
- (65) The specifying unit 42, for example, specifies the type of the item contained in the extracted basket region. Also for the specifying, for example, items may be detected from a captured image using an existing machine learning technique and a machine learning model that is generated by performing training such that the type of the detected item is output may be used.
- (66) The calculator 43, for example, calculates the number of the items stored in the basket based on the specified position of the outline of the item region. For example, the volume of items that is estimated by estimation of a volume of items like that described using FIGS. 9 to 12 is multiplied by a conversion factor, such as 20, thereby being converted into the number of items. More specifically, when the estimated volume of items is 0.3, the number of items is (conversion factor)×(volume of items)=20×0.3=6. Note that conversion into the number of items may be, for example, a conversion of (number of items)=f(volume of items) using a non-linear function f(x) or may be a conversion using an estimation system, such as a random forest using decision trees. The number of items may be calculated further based on the type of the item contained in the basket region, which is the type specified by the specifying unit 42.
- (67) For example, when the basket contains a large item, such as toilet paper or a set of boxes of tissue, there is a tendency that the number of extracted rectangular regions of the items is small and the area of one rectangular region is large. For this reason, a system like that described using FIGS. 9 to 12 may estimate a larger volume of items and resultantly estimate the larger number of items than the actual ones. Thus, also using information on the rectangles of the items in a visible area, for example, the number of or the size of the rectangles, for a conversion to the number of the items makes it possible to reduce errors in the estimation. In other words, the calculator 43 calculates the size of the rectangular regions of the items in the image and calculates a second number of the items extracted from the image. For example, based on the sizes of the regions and the second number of the items that are calculated and the positions of the outlines of the item regions that are specified by the specifying unit 42, the calculator 43 calculates the number of the items stored in the basket. More specifically, the calculator 43, for example, calculates the number of the items using Equations (2) to (4) below.

$$(68) \quad \begin{aligned} (\text{Number of items}) = & k1 \times (\text{estimated volume of items}) + \\ & k2 \times (\text{number of rectangular regions of respective items}) + \\ & k3 \times (\text{average of areas of rectangular regions of respective items}) \end{aligned} \quad (2)$$

$$\begin{aligned} (\text{Number of items}) = & k1 \times (\text{estimated volume of items}) + \\ & k2 \times (\text{number of rectangular regions of respective items}) + \\ & k3 \times (\text{average of areas of } N \text{ rectangular regions in descending order in size}) \end{aligned} \quad (3)$$

$$\begin{aligned}
& (\text{Number of items}) = k_1 \times (\text{estimated volume of items}) + \\
& k_2 \times (\text{number of rectangular regions of respective items}) + \\
& k_3 \times (\text{area of largest rectangular region}) + \\
& k_4 \times (\text{area of second largest rectangular region}) + \dots \text{Math.}
\end{aligned} \tag{4}$$

(69) In Equations (2) to (4), k_1 , k_2 and k_3 denote weight coefficients for the volume of items, the number of rectangular regions of the respective items, and an average of the areas of the rectangular regions of the respective items, respectively. The average of the areas of the rectangular regions of the respective items may be calculated using the areas of all the rectangular regions of the items that are extracted from the image as presented by Equation (2) or may be calculated using the areas of N rectangular regions in a descending order in size of area as presented by Equation (3). The average of the areas of the rectangular regions of the respective items may be calculated using the areas of N rectangular regions in the descending order in size of area as other explanatory variables as presented by Equation (4). The (number of rectangular regions of respective items) segment in Equations (2) to (4) may be “ $k_2 \times \max(0, M - \text{number of rectangular regions of respective items})$ ”. This is, for example, to make a correction only in the case where the number of the rectangular regions of the respective items is smaller than a given number M with a high possibility that there is a large item. More specifically, for example, when $M=4$ is set, $\max(0, a)$ is corrected and calculated as a if $a > 0$ and is calculated as 0 without correction if $a \leq 0$. The number of items may be calculated by, instead of such a linear regression, a method using a non-linear function or a method using decision trees.

(70) A method of calculating the number of items using a machine learning model will be described here. The number of items that are stored in a basket is specified by inputting at least one of the volume of the items, the areas of rectangular regions of the items, and the number of the rectangular regions of the items to a trained machine learning model.

(71) For example, the trained machine learning model is a machine learning model with a parameter of a neural network being changed such that an error between an output result that the neural network outputs when an explanatory variable that is the volume of items is input to the neural network and truth data that is a label of the number of the items is reduced.

(72) The trained machine learning model may be a machine learning model with a parameter of a neural network being changed such that an error between an output result that the neural network outputs when explanatory variables that are the volume of items, areas of bounding boxes, and the number of the bounding boxes are input to the neural network and truth data that is a label of the number of the items is reduced. The areas of the bounding boxes may be the areas of image regions of the items stored in the basket that are contained in the image and are specified from the rectangular regions of the respective items contained in the image. The number of the bounding boxes may be a total of the rectangular regions of the items that are specified from the number of the rectangular regions of the items.

(73) Back to FIG. 9, a method of specifying a volume of items using bounding boxes will be described here. For example, R-CNN or SSD is used for bounding boxes. R-CNN combines rectangular regions with features of a convolutional neural network and detects a subset of an area (region proposal). CNN features are then extracted from the region proposal. A support vector machine that is trained previously using the CNN features adjusts the bounding boxes of the region proposal.

(74) SSD is an algorithm of general object sensing using machine learning and determines bounding boxes from rectangular regions referred to as default boxes. Multiple default boxes in different sizes are layered on a single image and a prediction value is calculated per box. It is possible to predict, with respect to each of the default boxes, a position presenting how far the default box is from an object and how the default box is different from the object in size.

(75) The detector **41** specifies, from the image, a first bounding box of each item that surrounds the item and that is contained in the image. The detector **41** specifies a position of a lower border of a second bounding box.

(76) The specifying unit **42** specifies a position of an upper border of the first bounding box of each item contained in the image. The specifying unit **42** specifies a position corresponding to all the items stored in the basket from the specified positions of the upper borders of the respective items contained in the image. Furthermore, the specifying unit **42** specifies a second bounding box surrounding the basket from the acquired image.

(77) The calculator **43** specifies an item volume of the items that are stored in the basket based on the position corresponding to the entire items with respect to the position of the lower border. Furthermore, the calculator **43** further specifies the number of the items by inputting the item volume of the items to the trained machine learning model.

(78) Flow of Process

(79) A flow of a number-of-items estimating process that is executed by the information processing device **10** will be described. FIG. **13** is a flowchart illustrating an example of the flow of the number-of-items estimating process according to Example 1. The number-of-items estimating process illustrated in FIG. **13** may be executed, for example, at given intervals or every time a captured image is received from the camera device **200**.

(80) First of all, as illustrated in FIG. **13**, the information processing device **10**, for example, acquires a captured image of a basket that is captured by the camera device **200** from the image DB **31** (step **S101**). Note that, in the number-of-items estimating process illustrated in FIG. **13**, in order to process in real time the captured image, precisely, a monitoring video, that is captured by the camera device **200**, captured images are transmitted as needed from the camera device **200** and are stored in the image DB **31**.

(81) For example, using an existing object detection algorithm, the information processing device then detects a shopping basket from the captured image that is acquired at step **S101** (step **S102**). Note that, for detection of the shopping basket, a given region containing the shopping basket may be extracted by, for example, a bounding box, or the like.

(82) For example, using an existing object detection algorithm, the information processing device **10** detects an item that is stored in the shopping basket that is detected at step **S102** from the captured image that is acquired at step **S101** (step **S103**). Also for detection of the item, a given region containing the item may be extracted by, for example, a bounding box, or the like.

(83) The information processing device **10**, for example, specifies an outline of a region of a group of the items that are detected at step **S103** (step **S104**). This is, for example, as described using FIG. **9**, specifying a position of a top side of the outline of the item region **70** with a top side of the basket region **60** being set for 1 and a bottom side being set for 0. As described using FIG. **9**, the top side of the no-item region **80** may be specified as 0.

(84) The information processing device **10**, for example, calculates a volume of the items based on positions of the outlines of the region of the group of the items and the region without an item that are specified at step **S104** (step **S105**).

(85) The information processing device **10** then multiplies the volume of the items that is calculated at step **S105** by a conversion factor, such as 20, thereby converting the volume of the items into the number of items (step **S106**). After step **S106** is executed, the number-of-items estimating process illustrated in FIG. **13** ends.

(86) Using FIG. **14**, a different number-of-items estimating process from the number-of-items estimating process illustrated in FIG. **13** will be described next along a flow of the process. FIG. **14** is a flowchart illustrating another example of the flow of the number-of-items estimating process according to Example 1. Steps **S201** to **S205** in the number-of-items estimating process illustrated in FIG. **14** are the same as steps **S101** to **S105** in the number-of-items estimating process illustrated in FIG. **13**.

- (87) For example, as described using FIG. 12, using Equation (1), the information processing device **10** corrects the volume of the items that is calculated at step **S205** according to the position of the basket (step **S206**).
- (88) The information processing device **10**, for example, multiplies the volume of the items that is calculated at step **S206** by a conversion factor, such as 20, thereby converting the volume of the items into the number of items (step **S207**). After step **S207** is executed, the number-of-items estimating process illustrated in FIG. 14 ends.
- (89) Using FIG. 15, another different number-of-items estimating process from the number-of-items estimating processes illustrated in FIG. 13 and FIG. 14 will be described next along a flow of the process. FIG. 15 is a flowchart illustrating a still another example of the flow of the number-of-items estimating process according to Example 1. Steps **S301** to **S304** in the number-of-items estimating process illustrated in FIG. 15 are the same as steps **S101** to **S104** in the number-of-items estimating process illustrated in FIG. 13.
- (90) The information processing device **10** calculates the number of and size of the regions of the group of the items that are detected at step **S303** (step **S305**).
- (91) The information processing device **10**, for example, calculates a volume of the items based on positions of the outlines of the region of the group of the items and the region without an item that are specified at step **S304** (step **S306**).
- (92) The information processing device **10** then, for example, converts the volume of the items that is calculated at step **S306** into the number of the items using the number of and size of the regions of the group of the items that are calculated at step **S305** and any one of Equations (2) to (4) (step **S307**). After step **S307** is executed, the number-of-items estimating process illustrated in FIG. 15 ends.
- (93) Effects
- (94) As described above, the information processing device **10** acquires an image of a basket storing items that is captured, extracts a basket region of the basket that is contained in the image by analyzing the acquired image, extracts at least one item region contained in the extracted basket region by analyzing the acquired image, estimates an item volume of the items stored in the basket based on a position of an outline of the item region with respect to a position of the extracted basket region, and specifies the number of the item stored in the basket based on the specified item volume. When the number of items is calculated, it is possible to reduce the throughput of the information processing device **10**.
- (95) In this manner, the information processing device **10** extracts regions of a shopping basket and items from a captured image and calculates the number of the items based on a position of an outline of the item region with respect to the region of the shopping basket. Accordingly, the information processing device **10** is able to more accurately specify the number of items in a shopping basket in a system in which a customer personally scans items.
- (96) The information processing device **10** specifies, from the acquired image, a first bounding box with respect to each item that surrounds the item and that is contained in the image, specifies a position of an upper border of the first bounding box with respect to each item contained in the image, specifies a position corresponding to all the items stored in the basket from the specified position of the upper border with respect to each item contained in the image, specifies a second bounding box surrounding the basket from the acquired image, specifies a position of a lower border of the second bounding box, specifies an item volume of the items stored in the basket based on the position corresponding to all the items with respect to the position of the lower border, and outputs the number of the items stored in the basket based on the specified volume of the items.
- (97) Accordingly, the information processing device **10** is able to more accurately specify the number of items in a shopping basket in the system in which a customer personally scans items.
- (98) The information processing device **10** detects, from the acquired image, a first image region

with respect to each item contained in the image that surrounds the item, specifies each of a position of an upper border of the first image region with respect to each item contained in the image and the number of the first image regions, detects a second image region that surrounds the basket from the acquired image, specifies an item volume of the items stored in the basket based on a position of a border of the second image region and a position of the upper border of the first image region, specifies an area of an image region of the items stored in the basket that is contained in the image using the first image region with respect to each item, and specifies the number of the items stored in the basket by inputting the item volume, the area of the image region of the items, and the number of the first image regions into a trained machine learning model.

(99) Accordingly, the information processing device **10** is able to more accurately specify the number of items in a shopping basket in the system in which a customer personally scans items.

(100) A process of calculating the number of the items that is executed by the information processing device **10** includes a process of calculating a size of the region of the item in the image, which is a rectangular region, calculating a second number of the items that are extracted from the image, and calculating the number of the items stored in the basket based on the size of the region, the second number of the items, and the specified position of the outline.

(101) Accordingly, the information processing device **10** is able to more accurately specify the number of items in a shopping basket in the system in which a customer personally scans items.

(102) The information processing device **10** specifies a first self-checkout terminal device that a first person with the basket uses from a second image of an area containing self-checkout terminal devices, receives item information on items to be purchased from the first self-checkout terminal device, specifies a third number of the items to be purchased based on the item information, and determines whether there is an item that is not registered as an item to be purchased in the self-checkout terminal device based on the number of the items that is calculated based on the specified position of the outline and the third number.

(103) Accordingly, the information processing device **10** is able to more accurately specify the number of items in a shopping basket in the system in which a customer personally scans items.

(104) A process of receiving the item information that is executed by the information processing device **10** includes a process of receiving, from the first self-checkout terminal device, the item information that is generated in a way that an information processing terminal device that the first person uses scans and then registers barcodes or two-dimensional codes that are assigned to the items.

(105) Accordingly, the information processing device **10** is able to more accurately specify the number of items in a shopping basket in the system in which a customer personally scans items.

(106) The process of calculating the number of the items that is executed by the information processing device **10** includes a process of specifying a second number of the items that are extracted from the image, calculating an average of positions of top sides of outlines of the item regions in the image with respect to the extracted basket region, and calculating the number of the items stored in the basket by correcting the second number of the items based on the average.

(107) Accordingly, the information processing device **10** is able to more accurately specify the number of items in a shopping basket in the system in which a customer personally scans items.

(108) The information processing device **10** executes a process of specifying a personal item of a first person with the basket that is contained in the extracted basket region and the process of calculating the number of the items includes a process of calculating the number of the items stored in the basket excluding the specified personal item.

(109) Accordingly, the information processing device **10** is able to more accurately specify the number of items in a shopping basket in the system in which a customer personally scans items.

(110) A process of specifying a position of an outline that is executed by the information processing device **10** includes specifying a position of the outline by instance segmentation.

(111) Accordingly, the information processing device **10** is able to more accurately specify the

number of items in a shopping basket in the system in which a customer personally scans items.
(112) The information processing device **10** executes a process of specifying a type of the item contained in the extracted basket region and the process of calculating the number of the items includes a process of calculating the number of the items stored in the basket further based on the type of the item.

(113) Accordingly, the information processing device **10** is able to more accurately specify the number of items in a shopping basket in the system in which a customer personally scans items.

(114) The process of extracting the basket region that is executed by the information processing device **10** includes a process of, after executing rotational correction on the image, extracting the basket region in the image on which rotational correction has been performed.

(115) Accordingly, the information processing device **10** is able to more accurately specify the number of items in a shopping basket in the system in which a customer personally scans items.

(116) System

(117) The process procedure, control procedure, specific names, and information including various types of data and parameters that are presented in the above description and the drawings may be changed freely unless otherwise noted. Specific examples, distributions, and numerical values described in the example are examples only and they may be changed freely.

(118) Specific modes of distribution and integration of components of each device are not limited to those illustrated in the drawings. In other words, all or part of the components may be distributed or integrated functionally or physically in any unit according to various types of load and usage. Furthermore, all or given part of each processing function of each device can be realized by a CPU or a program that is analyzed and executed by the CPU or can be realized as hardware according to a wired logic.

(119) Hardware

(120) FIG. **16** is a diagram illustrating an example of a hardware configuration of the information processing device **10**. As illustrated in FIG. **16**, the information processing device **10** includes a communication interface **10a**, a hard disk drive (HDD) **10b**, a memory **10c**, and a processor **10d**. The units illustrated in FIG. **16** are connected mutually via a bus, or the like.

(121) The communication interface **10a** is a network interface card, or the like, and communicates with another server. The HDD **10b** stores a program that runs the functions illustrated in FIG. **5** and a DB.

(122) The processor **10d** is a hardware circuit that reads the program for executing the same process as that of each of the processing units illustrated in FIG. **5** from the HDD **10b**, or the like, and loads the program in the memory **10c**, thereby running the process that implements each of the functions illustrated in FIG. **5**, etc. In other words, the process executes implements the same function as that of each processing unit that the information processing device **10** includes. Specifically, the processor **10d** reads the program with the same functions as those of the detector **41**, the specifying unit **42**, the calculator **43**, the notification unit **44**, etc., from the HDD **10b**, or the like. The processor **10d** executes the process of executing the same process as that performed by the specifying unit **42**, etc.

(123) As described above, by reading and executing the program for executing the same process as that of each of the processing units illustrated in FIG. **5**, the information processing device **10** runs as an information processing device that executes an operation control process. The information processing device **10** is also able to realize the same functions as those of the above-described example by reading the program from a recording medium using a medium reading device and executing the read program. Programs according to other examples are not limited to being executed by the information processing device **10**. For example, the present embodiment may be similarly applied to the case where another information processing device executes the program or the information processing device **10** and another information processing device execute the program cooperatively.

(124) The program for executing the same process as that of each of the processing units illustrated in FIG. 5 can be distributed via a network, such as the Internet. The program is recorded in a computer-readable recording medium, such as a hard disk, a flexible disk (FD), a CD-ROM, a magneto-optical disk (MO), or a digital versatile disc (DVD) and is read by a computer from the recording medium, so that the program can be executed.

(125) FIG. 17 is a diagram illustrating an example of a hardware configuration of an information processing terminal device **1000**. The user terminal device **100** and the store staff terminal device **300** may have the same configuration and therefore the user terminal device **100** and the store staff terminal device **300** are collectively referred to as the information processing terminal device **1000** and the information processing terminal device **1000** is illustrated as an example of a hardware configuration of the user terminal device **100** and the store staff terminal device **300**. As illustrated in FIG. 17, the information processing terminal device **1000** includes a communication interface **1000a**, a HDD **1000b**, a memory **1000c**, a processor **1000d**, an input device **1000e**, and a display device **1000f**. The units illustrated in FIG. 17 are connected mutually via a bus, or the like.

(126) The communication interface **1000a** is a network interface card, or the like, and communicates with another information processing device. The HDD **1000b** stores a program that runs each function of the information processing terminal device **1000** and data.

(127) The processor **1000d** is a hardware circuit that reads the program for executing a process of each function of the information processing terminal device **1000** from the HDD **1000b**, or the like, and loads the program in the memory **1000c**, thereby running a process that implements each function of the information processing terminal device **1000**. In other words, the process implements the same function as that of each processing unit that the information processing terminal device **1000** includes.

(128) As described above, by reading and executing the program for executing the process of each function of the information processing terminal device **1000**, the information processing terminal device **1000** runs as an information processing device that executes an operation control process. The information processing terminal device **1000** is also able to realize each function of the information processing terminal device **1000** by reading the program from a recording medium using a medium reading device and executing the read program. Programs according to other examples are not limited to being executed by the information processing terminal device **1000**. For example, the present embodiment may be similarly applied to the case where another information processing device executes the program or the information processing terminal device **1000** and another information processing device execute the program cooperatively.

(129) The program for executing the process of each function of the information processing terminal device **1000** can be distributed via a network, such as the Internet. The program is recorded in a computer-readable recording medium, such as a hard disk, a FD, a CD-ROM, a MO, or a DVD and is read by a computer from the recording medium, so that the program can be executed.

(130) The input device **1000e** senses various input operations of a user, such as an input operation on the program that is executed by the processor **1000d**. The input operation includes, for example, a touch operation and insertion of an earphone terminal to the information processing terminal device **1000**. The touch operation refers to various contact operations on the display device **1000f**, for example, a tap, a double tap, a swipe, and a pinch. The touch operation includes, for example, a motion of moving an object, such as a finger, close to the display device **1000f**. The input device **1000e** may be, for example, a button, a touch panel, or a proximity sensor.

(131) The display device **1000f** displays various types of visual information according to control performed by the processor **1000d**. The display device **1000f** may be a liquid crystal display (LCD) or an organic light emitting diode (OLED) that is what is referred to as an organic electroluminescence (EL) display.

(132) FIG. 18 is a diagram illustrating an example of a hardware configuration of the self-checkout

terminal device **400**. As illustrated in FIG. **18**, the self-checkout terminal device **400** includes a communication interface **400a**, a HDD **400b**, a memory **400c**, a processor **400d**, an input device **400e**, and an output device **400f**. The units illustrated in FIG. **18** are connected mutually via a bus, or the like.

(133) The communication interface **400a** is a network interface card, or the like, and communicates with another information processing device. The HDD **400b** stores a program that runs each function of the self-checkout terminal device **400** and data.

(134) The processor **400d** is a hardware circuit that reads the program for executing a process of each function of the self-checkout terminal device **400** from the HDD **400b**, or the like, and loads the program in the memory **400c**, thereby running a process that implements each function of the self-checkout terminal device **400**. In other words, the process implements the same function as that of each processing unit that the self-checkout terminal device **400** includes.

(135) As described above, by reading and executing the program for executing the process of each function of the self-checkout terminal device **400**, the self-checkout terminal device **400** runs as an information processing device that executes an operation control process. The self-checkout terminal device **400** is also able to implement each function of the self-checkout terminal device **400** by reading the program from a recording medium using a medium reading device and executing the read program. Programs according to other examples are not limited to being executed by the self-checkout terminal device **400**. For example, the present embodiment may be similarly applied to the case where another information processing device executes the program or the self-checkout terminal device **400** and another information processing device execute the program cooperatively.

(136) The program for executing the process of each function of the self-checkout terminal device **400** can be distributed via a network, such as the Internet. The program is recorded in a computer-readable recording medium, such as a hard disk, a FD, a CD-ROM, a MO, or a DVD and is read by a computer from the recording medium, so that the program can be executed.

(137) The input device **400e** senses various input operations of a user, such as an input operation on the program that is executed by the processor **400d**. The input operation includes, for example, a touch operation. In the case of the touch operation, the self-checkout terminal device **400** further includes a display device and the input operation that is sensed by the input device **400e** may be a touch operation on the display device. The input device **400e** may be, for example, a button, a touch panel, or a proximity sensor.

(138) The output device **400f** outputs data that is output from the program that is executed by the processor **400d** via an external device that is connected to the self-checkout terminal device **400**, for example, an external display device. Note that, when the self-checkout terminal device **400** includes a display device, the self-checkout terminal device **400** need not include the output device **400f**.

(139) According to an aspect, in a system in which a customer personally scans items, it is possible to more accurately specify the number of items in a shopping basket.

(140) All examples and conditional language recited herein are intended for pedagogical purposes of aiding the reader in understanding the invention and the concepts contributed by the inventor to further the art, and are not to be construed as limitations to such specifically recited examples and conditions, nor does the organization of such examples in the specification relate to a showing of the superiority and inferiority of the invention. Although the embodiment of the present invention have been described in detail, it should be understood that the various changes, substitutions, and alterations could be made hereto without departing from the spirit and scope of the invention.

Claims

1. A non-transitory computer-readable recording medium having stored therein an information processing program that causes a computer to execute a process comprising: acquiring an image of an object storing items that is captured; extracting an object region of the object contained in the image by analyzing the image acquired; extracting one or more item regions contained in the extracted object region by analyzing the image acquired; estimating an item volume of the items in the object based on a position of an outline of the item region with respect to a position of the extracted object region; and specifying a number of items in the object based on the specified item volume, wherein the process further includes: specifying, from the acquired image, a first bounding box with respect to each item that surrounds the item and that is contained in the image; specifying a position of an upper border of the first bounding box with respect to each item contained in the image; specifying a position corresponding to all the items in the object from the position of the upper border with respect to each item contained in the image; specifying a second bounding box that surrounds the object from the acquired image; specifying a position of a lower border of the second bounding box; specifying an item volume of the items in the object based on the position corresponding to all the items with respect to the position of the lower border; and outputting the number of the items in the object based on the specified volume of the items.
2. The non-transitory computer-readable recording medium having stored therein according to claim 1, wherein the program further causes the computer to execute detecting, from the acquired image, a first image region with respect to each item that surrounds the item and that is contained in the image; specifying each of a position of an upper border of the first image region with respect to each item that is contained in the image and the number of the first image regions; detecting a second image region that surrounds the object from the acquired image; specifying an item volume of the items in the object based on a position of a border of the second image region and the position of the upper border of the first image region; specifying an area of an image region of the items in the object that is contained in the image using the first image region with respect to each item; and specifying the number of the items in the object by inputting the item volume, the area of the image region of the items, and the number of the first image regions into a trained machine learning model.
3. The non-transitory computer-readable recording medium having stored therein according to claim 1, wherein the specifying the number of the items includes calculating a size of the region of the item in the image, which is a rectangular region, calculating a second number of the items extracted from the image, and specifying the number of the items in the object based on the size of the region, the second number of the items, and the specified position of the outline.
4. The non-transitory computer-readable recording medium having stored therein according to claim 1, wherein the program further causes the computer to execute specifying a first self-checkout terminal that a first person with the object uses from a second image of an area containing self-checkout terminals, receiving item information on items to be purchased from the first self-checkout terminal, specifying a third number of the items to be purchased based on the item information, and determining whether there is the item that is not registered as the item to be purchased in the self-checkout terminal based on the number of the items that is calculated based on the specified position of the outline and the third number.
5. The non-transitory computer-readable recording medium having stored therein according to claim 4, wherein the receiving the item information includes receiving, from the first self-checkout terminal, the item information that is generated in a way that an information processor that the first person uses scans and then registers barcodes or two-dimensional codes that are assigned to the items.
6. The non-transitory computer-readable recording medium having stored therein according to claim 1, wherein the estimating the item volume includes calculating an average of positions of top sides of outlines of the item regions in the image with respect to the extracted object region, and

estimating the item volume by correcting the average based on the object position.

7. The non-transitory computer-readable recording medium having stored therein according to claim 1, wherein the program further causes the computer to execute specifying a personal item of a first person with the object that is contained in the extracted object region, and the specifying the number of the items includes specifying the number of the items in the object excluding the specified personal item.

8. The non-transitory computer-readable recording medium having stored therein according to claim 1, wherein the program further causes the computer to execute specifying a position of the outline by instance segmentation.

9. The non-transitory computer-readable recording medium having stored therein according to claim 1, wherein the program further causes the computer to execute specifying a type of the item contained in the extracted object region, and the calculating the number of the items includes calculating the number of the items in the object further based on the type of the item.

10. The non-transitory computer-readable recording medium having stored therein according to claim 1, wherein the extracting the object region includes, after executing rotational correction on the image, extracting the object region in the image on which rotational correction has been performed.

11. An information processing method executed by a computer, the method comprising: acquiring an image of an object storing items that is captured; extracting an object region of the object contained in the image by analyzing the image acquired; extracting one or more item regions contained in the extracted object region by analyzing the image acquired; estimating an item volume of the items in the object based on a position of an outline of the item region with respect to a position of the extracted object region; and specifying a number of items in the object based on the specified item volume, wherein the method further includes: specifying, from the acquired image, a first bounding box with respect to each item that surrounds the item and that is contained in the image; specifying a position of an upper border of the first bounding box with respect to each item contained in the image; specifying a position corresponding to all the items in the object from the position of the upper border with respect to each item contained in the image; specifying a second bounding box that surrounds the object from the acquired image; specifying a position of a lower border of the second bounding box; specifying an item volume of the items stored in the object based on the position corresponding to all the items with respect to the position of the lower border; and outputting the number of the items in the object based on the specified volume of the items.

12. The information processing method according to claim 11, further comprising: detecting, from the acquired image, a first image region with respect to each item that surrounds the item and that is contained in the image; specifying each of a position of an upper border of the first image region with respect to each item that is contained in the image and the number of the first image regions; detecting a second image region that surrounds the object from the acquired image; specifying an item volume of the items in the object based on a position of a border of the second image region and the position of the upper border of the first image region; specifying an area of an image region of the items in the object that is contained in the image using the first image region with respect to each item; and specifying the number of the items in the object by inputting the item volume, the area of the image region of the items, and the number of the first image regions into a trained machine learning model.

13. The information processing method according to claim 11, wherein the specifying the number of the items includes calculating a size of the region of the item in the image, which is a rectangular region, calculating a second number of the items extracted from the image, and specifying the number of the items in the object based on the size of the region, the second number of the items, and the specified position of the outline.

14. The information processing method according to claim 11, further comprising: specifying a first

self-checkout terminal that a first person with the object uses from a second image of an area containing self-checkout terminals, receiving item information on items to be purchased from the first self-checkout terminal, specifies a third number of the items to be purchased based on the item information, and determining whether there is the item that is not registered as the item to be purchased in the self-checkout terminal based on the number of the items that is calculated based on the specified position of the outline and the third number.

15. An information processing device, comprising: a memory; and a processor coupled to the memory and the processor configured to: acquire an image of an object storing items that is captured; extract an object region of the object contained in the image by analyzing the image acquired; extract one or more item regions contained in the extracted object region by analyzing the image acquired; estimate an item volume of the items in the object based on a position of an outline of the item region with respect to a position of the extracted object region; and specify a number of items in the object based on the specified item volume, wherein the processor is further configured to: specify, from the acquired image, a first bounding box with respect to each item that surrounds the item and that is contained in the image; specify a position of an upper border of the first bounding box with respect to each item contained in the image; specify a position corresponding to all the items in the object from the position of the upper border with respect to each item contained in the image; specify a second bounding box that surrounds the object from the acquired image; specify a position of a lower border of the second bounding box; specify an item volume of the items in the object based on the position corresponding to all the items with respect to the position of the lower border; and output the number of the items in the object based on the specified volume of the items.

16. The information processing device according to claim 15, wherein the processor configured to detect, from the acquired image, a first image region with respect to each item that surrounds the item and that is contained in the image; specify each of a position of an upper border of the first image region with respect to each item that is contained in the image and the number of the first image regions; detect a second image region that surrounds the object from the acquired image; specify an item volume of the items in the object based on a position of a border of the second image region and the position of the upper border of the first image region; specify an area of an image region of the items in the object that is contained in the image using the first image region with respect to each item; and specify the number of the items in the object by inputting the item volume, the area of the image region of the items, and the number of the first image regions into a trained machine learning model.

17. The information processing device according to claim 15, wherein the specifying the number of the items includes calculating a size of the region of the item in the image, which is a rectangular region, calculating a second number of the items extracted from the image, and specifying the number of the items in the object based on the size of the region, the second number of the items, and the specified position of the outline.

18. The information processing device according to claim 15, wherein the processor configured to specify a first self-checkout terminal that a first person with the object uses from a second image of an area containing self-checkout terminals; receive item information on items to be purchased from the first self-checkout terminal; specify a third number of the items to be purchased based on the item information; and determine whether there is the item that is not registered as the item to be purchased in the self-checkout terminal based on the number of the items that is calculated based on the specified position of the outline and the third number.

19. The information processing device according to claim 18, wherein the receiving the item information includes receiving, from the first self-checkout terminal, the item information that is generated in a way that an information processing terminal that the first person uses scans and then registers barcodes or two-dimensional codes that are assigned to the items.

20. The information processing device according to claim 15, wherein the estimating the item

volume includes calculating an average of positions of top sides of outlines of the item regions in the image with respect to the extracted object region, and estimating the item volume by correcting the average based on the object position.
